

Goo-y: A Robot's Shape-Changing Capability for Fast, Non-Anthropomorphic Communication with People (and Animals)

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Abstract

Effective interaction with robots relies heavily on their ability to convey their purpose and intentions. While people are familiar with communication methods like (human) speech, this may lead to unnecessary anthropomorphisation or deception. On the other hand, robot-specific signals such as movements or colours are often difficult to interpret. Drawing inspiration from interspecies communication and animation principles offers a broader design space for creating expressive, easily understood signals through *shape-changing capabilities*. We detail the development of a shape-changing capability for a robot to support non-anthropomorphic, visceral communication. Starting with a virtual reality study to gauge people's responses to various shape-changing behaviours, followed by developing a full-scale version suitable for standard robotic platforms and in-situ experimentation. This low-cost, easy-to-build capability opens new possibilities for researchers in human-robot interaction by making shape-changing technology accessible to the CHI research community.

CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design**; • **Computer systems organization** → **Robotics**.

Keywords

robots, appearance, shape-change, multimodal interaction design, communication, expectations

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1 Introduction

To date, robots have been largely deployed at scale in controlled industrial environments, such as factories or warehouses. However, to help solve a range of societal challenges (e.g., a lack of trained personnel), robots are more and more developed to operate in everyday workplaces such as restaurants, hospitals and shops, where robots share workspaces with people. The work reported in this paper is carried out in the context of service robots in work settings, where they function in close proximity to people. We have situated this in environments such as in healthcare and the agriculture-food industry. The people in these work environments are not uniform; in a hospital, staff, visitors and patients all have different needs and expectations when interacting or encountering a service robot. And in agricultural settings, encounters with animals may be considered. To ensure the robot's signals are clear, acceptable, and effective, robots need a range of expressive modalities. This paper presents the design of a robot that uses shape-change as a novel modality for communication. Ongoing research is assessing whether a robot's shape change can provide immediate, visceral communication to diverse users so that it is instantly clear what the purpose of the robot is and what it is doing.

Designers often use *anthropomorphic features* in robots to make their communication more intuitive for people. After all, we are wired to understand other people, so robots that communicate in a manner similar to people leverage that sensemaking process [28]. However, this approach has significant drawbacks. It risks reinforcing human biases, such as racial and gender stereotypes [2, 16, 34],

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and can lead to overestimating the robot's abilities, potentially causing confusion or safety risks [8, 19, 29]. Anthropomorphic communication can also create ethical concerns, as it may mislead people about the robot's true capabilities or intentions [5, 24, 30]. Given these risks of anthropomorphic robot design, designers should exercise caution applying it. [26]. The British Standards Institution also advises against "unnecessary anthropomorphization" and "deception due to the behaviour and/or appearance of the robot and ensure transparency of its robotic nature" [4, pp. 3,]. In hospital or agricultural settings, where a robot's primary role is generally not interacting with people but carrying out physical tasks, we argue that humanlike behaviour such as speaking or facial expressions creates further incongruency and does not effectively support the robot's interactions with people it encounters.

While *non-anthropomorphic design* of robots and their behaviours may address concerns outlined above, designing robot-specific signals that are clear and transfer information effectively, is difficult. People are not accustomed to interpreting non-human signals, and may find them hard to understand [10]. To address the challenge of designing non-anthropomorphic modes of communication, shape-changing may be a mode of communication that offers a wide range of expressivity. Shape-change is used by animals to communicate [3] (e.g., the pufferfish [23, 33]), and by animators to bring characters to life in animations [21, 32]. As shown by numerous shape-changing interfaces, shape-change can be used for the design of highly expressive, legible, and non-disruptive, non-anthropomorphic signals [1, 6, 12, 13, 25]. The design space for shape-change for robot communication remains relatively unexplored [13], where most studies do not go beyond small-scale prototypes of the shape-changing capability [e.g., 7, 11, 15, 17, 18, 20, 27].

We believe that shape-change may be an excellent modality for non-anthropomorphic communication for service robots in the context of work: the shape-changing signals have the potential to be immediately understood, may transcend communication between species, can be visible from various viewpoints, and allow non-invasive communication (in contrast to using sound) as to not disturb tranquility in for instance hospital spaces or disturb people or animals at work in greenhouses or livestock farms. The design space for shape-change as a non-anthropomorphic way for robots to communicate with people is large. This design space also remains relatively unexplored, as current shape-changing robots were aimed mostly at movement or conveying emotions, but not other types of communication. Moreover, in-situ studies of shape-changing robots and the effect they have on people are largely missing. Current prototypes are ill-equipped for such studies, as they are not full-functioning robots, but instead are used to explore shape-changing capabilities in (university) lab settings. To enable in-situ studies and expand the numerous ways shape-changing can be deployed by robots for communication, we developed a functional, mobile basic robot that can change shape in terms of volume and form.

In this paper, we report the development of the Goo-y robot: a non-anthropomorphic robot that features a soft silicon shell uses inflation to modify its volume, and can change its body in all directions. It is mounted on a mobile base that allows it to move and is designed to be able to operate in various work settings such as hospitals, office buildings and even livestock-stables. We describe the shape-changing capability, and an exploratory study in virtual

reality (VR) that investigated the perception of the shape-changing behaviours.

2 The Shape-Changing Capability in Virtual Reality

A pilot study was carried out to explore the communicative potential of the shape-changing capability and gather feedback on how people experienced the shape-changing expressions. The study took place at the University Hospital of Zürich and involved five participants who were hospital staff members. The study was reviewed by the ethical committee of the University of Twente (reference number: 240454).

2.1 Experiment design

Participants experienced the shape-changing capability in VR during sessions of approximately one hour each. Participants were exposed to distinct communicative behaviours of the robot in the VR environment, where the intention was given shape with a static embodiment, or the shape-changing embodiment. For the latter, the upper body and head were rigid and the shell was static. In both the shape-changing and static embodiments, the robot adopted eye expressions, consequential sound, base LED colours and movements. While the shape-changing embodiments showed changes in the volume (height, width and rotation) and form (roundness or squareness of the shell) (see Figure 1). We tested five communicative behaviours, each with two designs, where the scene was the same.

2.2 Procedure and measures

Participants were asked to observe and occasionally interact with the robot within the VR setting, where they encountered the robot navigating in front and around them. After these interactions, a window would appear in the VR environment where they could see the experimenter and the outside world. This allowed the participant to have a face-to-face semi-structured interview with the experimenter to gauge their perceptions of the robot's behaviour. The questions focused on describing what they had just experienced, the robot's behaviours, and their feelings and attitudes towards the robot. Participants also engaged with the interviewer in a discussion on how the robot's behaviour might impact their daily workflow, their sensemaking process of how they interpreted the robot's behaviour and any expectations or feelings they may have had on what the robot should have expressed for their improved work-experience.

2.3 VR environment and behaviour design

The VR environment aims to explore how people experience sharing (virtual) space with a robot, while the robot is driven by its ROS2 navigation components. The robot in VR is simulated fully — it features virtual sensors, such as a virtual IMU, based on the rigid body information from the Unity3D physics engine, and virtual lasers using a simple raycasting method. The motors/drivetrain is also simulated using two torque-driven wheel colliders with a PID controller to translate target velocity from Twist messages torque.

The behaviours we designed for human-robot communication in hospital hallways that involve shape-changing are given below.

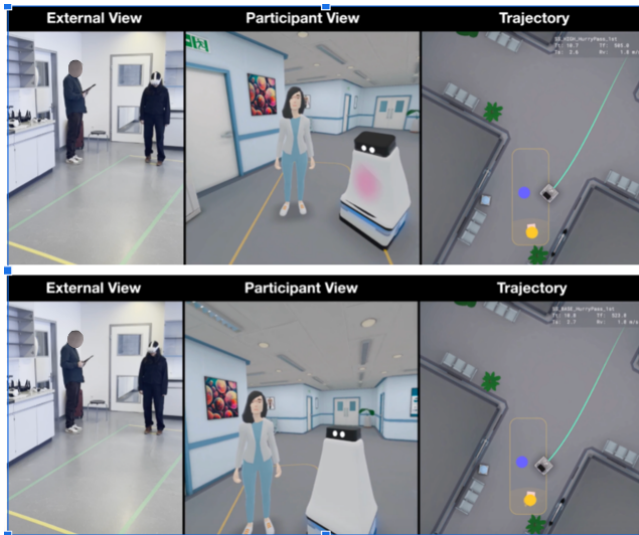


Figure 1: Example scenarios of the shape-changing behaviours (top) or static embodiment (bottom).

For the static embodiment, the intentions were the same, but the design did only use eye expressions, consequential sound, LED colours on the base, and movement of the base.

- **Intention to start moving.** Reduce volume, lean back, then lean forward, return to normal volume, and accelerate.
- **Coming to a stop.** Reduce volume, lean forward, then lean back to the centre and return to normal volume as it reaches the destination.
- **Robot is “in a hurry” (urgency).** Lean forward while moving and showing a diffused red light through its skin.
- **Intention to pass through a tight spot between people.** Reduce volume and squash to make itself smaller while passing through.
- **Reject a bid for interaction.** Expand in volume, lean forward, and a diffused red light shining through the skin until past the person.

2.4 Results and discussion

Participants found the shape-changing robot particularly effective in creating a sense of awareness and (psychological) safety. They felt that the robot adjusted its movements to accommodate their presence in the hallways, which made its behaviour appear intentional and reassuring. This sense of safety was attributed to the robot’s ability to communicate navigational intent more dynamically than its static counterpart.

The shape-changing robot also elicited stronger and more engaging reactions. Participants frequently laughed, made louder comments, and responded visibly through facial expressions and body language. Movements such as inflating, deflating, and shifting between rounded and square shapes, as well as rhythmic up-and-down motions, were described as cheerful and lively. One participant mentioned that it seemed the shape-changing robot was enjoying its work and could be left alone. To dig into this interpretation we

discussed with participants the analogy with a plumber or a bricklayer, singing or whistling while they are doing their job. They are busy and active, it is also a signal that it is not the right moment to interrupt them as they seem ‘in the flow’.

In contrast, the static robot, which relied on colour changes, sound, and directional cues like utilising eye gaze to signal its intended path, was less noticeable and less intuitive. Participants often took longer to interpret the static robot’s intent, and its communication style failed to elicit the same level of engagement or immediacy as the shape-changing robot.

The shape-changing behaviours also prompted rich interpretations, as participants associated the robot’s movements with familiar human or animal behaviours. For instance, when the robot scrunched down and tilted forward without providing additional cues, it was perceived as “stressed” or “in a hurry”. Participants drew comparisons to observing animals in nature, such as squirrels gathering food or livestock grazing, absorbed in their tasks and indifferent to human observers. These analogies highlighted how the robot’s movements evoked natural and relatable impressions, even in the absence of explicit communication.

Participants emphasized the importance of visibility in the design of shape-changing robots. Large, noticeable shape changes were particularly effective in capturing attention and facilitating navigation in shared spaces like hospital hallways. For medical personnel, it was sufficient for the robot to demonstrate awareness of their presence and indicate its intended movements. However, participants noted that patients or visitors might respond differently to such behaviours, suggesting that further research is needed to explore how diverse user groups interpret and interact with shape-changing robots.

3 The Goo-y Shape-Changing Robot

The final, full-sized prototype is an advanced robotic character designed with fluid and organic movement capabilities (see the supplementary files for a video of its shape-changing capabilities and Figures 2 and 4). The robot features five axes of motion. Each axis offers a distinct movement: move the body forward and backward or body side to side, move the head upwards or downwards, or side to side, and contract or extend the body length. The body-axis is combined with inflation capabilities allowing a change of approximately 30 cm in circumference. The robot was built on a generic square-shaped robot base platform that is approximately 52 cm in width. However, the capability can be constructed on most robot base platforms available. The height of the shape-changing frame together with the base platform is approximately 108 cm (Figure 5).

Motors and Control System. The robot’s movements are powered by five *Mini Cheetah*-style motors each with an 8:1 reductor, known for their high torque and precision. These integrated drive motors are daisy-chained, which allows for an efficient distribution of power and data. The motor control data is supplied by an *ESP32*, which runs a customized version of *Bottango*—an open-source robotic control software designed for animating robots with smooth, lifelike movements.

Inflation System. The shape-changing robot also includes an air blower, utilising a standard *EDF* (Electric Ducted Fan) paired with an *ESC* (Electronic Speed Controller) for controlling the airflow. The



Figure 2: The Goo-y Shape-Changing Robot.

ESC is controlled by the *ESP32* via *PWM* (Pulse Width Modulation), allowing dynamic control of the blower's speed. This dynamic speed allows the robot to deflate and inflate quickly.

Structural Design. The frame of the robot is made from *aluminium extrusion*, providing both strength and flexibility for adjustments during assembly and maintenance. The structure is connected by 10mm *POM* (Polyoxymethylene) parts, laser-cut for precision. Although the original design called for 6mm laser-cut metal components, the *POM* parts were chosen for ease of machining, while maintaining structural integrity.

Skin and Aesthetic. The robot's external skin is made with flexible silicone, providing a pleasing tactile appearance. The skin is crafted from *Dragonskin* silicone, produced by *Smooth-On*, and utilizes a specific mixture of 100 parts *Part A*, 50 parts *Slacker*, and 100 parts *Part B*. This blend ensures a soft, malleable texture that allows for organic skin movement. The skin has been tinted white using a silicone pigment and cast in a custom wooden mould,



Figure 3: The Goo-y robot's silicon shell.



Figure 4: The structural design of the Goo-y robot.

treated with a protective paint coating to prevent the silicone from adhering during the moulding process.

Base. The Goo-y Robot is mounted on a robust robot base manufactured by *IDMind*, a robotics company based in Portugal. This base is an integral component of the system, as it includes high-performance motors and batteries that provide mobility and stability to the robot. Additionally, the base houses an onboard computer responsible for controlling the robot's spatial movements, it provides navigation and positioning.

To support the research community in having access to the shape-changing capability, a *STEP* package was developed which contains all the needed information for the robot to be replicated (see supplementary materials).

4 Discussion and Conclusion

In this paper, we reported on a novel interactive robot that utilises shape-change for non-anthropomorphic human-robot communication: the Goo-y Shape-Changing Robot. The robot is functional, mobile, and to scale, enabling in situ research to investigate people's responses to and interactions with a shape-changing robot. We showed various ways the robot can use shape-change for communication with people in hospital setting. These communicative behaviours were implemented in a digital twin of the robot, for which we conducted an exploratory evaluation in VR. The responses that we got led us to believe that shape-changing communication can be a highly expressive and legible form of non-anthropomorphic human-robot communication.

The Goo-y robot is not without limitations. For the hospital delivery robot use cases, a soft appearance may lead specific audiences at the hospital, for instance, curious children in the children's ward, to start interacting with the robot or wanting to hug or touch it. This would be detrimental to the primary role of the robot, processing and handling goods within the hospital, and will need extra design or functionality to resolve such interactions adequately. This warrants further study through an in-situ evaluation of the robot. Another limiting factor is that the Goo-y robot is a prototype that can be used for studying how shape-change capabilities for in situ

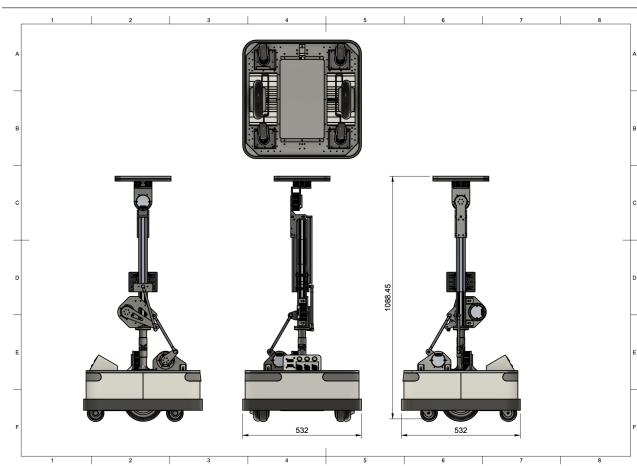


Figure 5: A render of the base and structure of the Goo-y robot.

experiments. It will need further design to incorporate other types of functionality, such as drawers for delivery tasks, or arms for carrying out manipulation tasks which will impact how the shape-changing is given form.

In future work, we plan on exploring how shape-change in texture can be used for communication. To further enhance the shape-changing capabilities of the Goo-y robot, combining the use of lights with shape-changing capabilities could be a promising avenue for non-anthropomorphic multimodal signals. The lights could be used to display markings, colours, or dynamic patterns to further enhance its non-anthropomorphic communication. In interspecific communication, colouration [i.e., aposematism, 9] can signal danger, such as poison frogs whose bright colours warn of their toxicity [22, 31]. Colouration can also be dynamically changed to create dynamic patterns. For instance, cephalopods can modify the form and texture by altering the appearance of their skin using specialised cells to create such (multimodal) dynamic patterns [14].

The shape-changing capability introduced in this paper opens up numerous possibilities for future research, particularly exploring novel methods of non-anthropomorphic communication for robots. This approach helps address societal and ethical concerns related to anthropomorphic design by providing a highly expressive alternative mode of interaction. By sharing the design of the Goo-y robot, we aim to contribute to the research community, offering insights into this emerging form of robot communication and encouraging further exploration of shape-changing as a communication modality.

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